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INVESTIGATIONS OF NUCLEAR STRUCTURE IN MASS-9 SYSTEMS USING NEUTRON SPECTROSCOPY

By

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TABLE OF CONTENTS

ABSTRACT

The mass 9 nuclear systems represents a significant deviation from the IMME, prompting investigation over non-trivial effects arising from isospin and nuclear structure considerations.

CHAPTER 1

INTRODUCTION

1.1 Motivation

One of the great triumphs of the past century of physics has been the illumination of the internal structure of the atom. It is well known that atomic nuclei are comprised of protons and neutrons and are held together by the strong force. It is also well-known that the interaction of these nuclei are constrained by quantum numbers such as isospin.

Isospin is a good quantum number when describing strongly interacting nucleons. Under the formalism describing isopsin, protons and neutrons both have total isospin $T = \frac{3}{2}$. Similar to intrinsic spin, the isospin z-projection T_z takes the values $\mp\frac{1}{2}$ for protons and neutrons respectively. Historically, isospin was derived from W. Heisenberg's attempt to treat both protons and neutrons as two states of the same particle, which was expanded upon by E. Wigner and given the name 'isotopic spin' as a means of listing the emergent symmetries of strongly-interacting nuclear systems [**wigner1937consequences**]. In particular, this 'isotopic spin', or isospin, plays a crucial role in defining Isobaric Analog States (IAS), which are states with equivalent isospin T with differing T_z values. From just the strong interaction, IAS would have equivalent energy levels. Due to the presence of Coulombinteractions from proton-proton repulsion, the isospin symmetry between IAS is broken, giving rise to differences in energy levels [**zelevinsky2017physics**]. The contributions to these energy shifts arise from various admixtures from interactions within the nucleus.

An example is the Thomas-Ehrman shift, an additional energy shift between the energy states in mirror nuclei from higher-order Coulomb interactions. Initially proposed by J. Ehrman and R. Thomas to describe the displacement in the first excited states between C^{13} and N^{13}[**ehrman1951displacement**] [**thomas1952analysis**], this shift is attributed to the interaction between nuclei structured as a core and valence nucleon. This extra shift supresses energy levels in proton-rich nuclei and is usually more pronounced in states with a weak centrifugal barrier. In particular, the mirror-pair ^9B and ^9Be offer a system which may exhibit a non-trivial TE contribution.

Since ^9B and ^9Be constitute an isopsin doublet, there should be a correspondence between the energy levels in both nuclei, accounting for Coulomb effects and proton/neutron mass differences. The $\frac{1}{2}+$ state in ^9Be is well measured with an excitation energy of 1.684 MeV [**Kuechler**], in contrast to the corresponding state in ^9B. There is evidence that both nuclei exhibit a three-body nature, with ^9Be being a Borromean system [**von-Oertzen**]. This indicates that both nuclei can be accurately represented as a core and valence nucleon systems.

Various measurements of the $\frac{1}{2}+$ state in ^9B have been attempted. After the first official report in 1955 [**PhysRev.100.91**], other authors have attempted to characterize this state. Historically, two different values have been proposed for this state. In 1962, G. Symons measured the state at 1.7(2) MeV[**Symons**], which would be contrasted with M. Tiede, who found that the state seemingly split into values of 0.8 and 1.6 MeV [**tiedde1995**]. Unsuccessful searches have continued; however, certain promising developments have recently occured. An solution to the apparent bifurcation of the $\frac{1}{2}+$ state was determined by Brooks et al, who proposed a "ghost peak" due to leakage of the ground state, appearing near 0.8 MeV. This would be followed by another paper which a resonance was characterized with an energy of $1.83(5)MeV$ and intrinsic width of $740^{110}_{88}keV$. However, this characterization did not could not unambiguously confirm this state as the $\frac{1}{2}+$ state since angular distributions could not be extracted[**5rny-7zbk**]. Further work is required to properly characterize this state.

In characterizing the $\frac{1}{2}+$ of ^9B, the $(^3He, n)$ reaction has not been properly utilized. There have been studies into the structure of ^9B using this reaction, starting in 1965 with Dietrich's investigation using time-of-flight spectroscopy using a plastic scintillator for neutron measurement. Highly excited states in ^9B where measured at 12.06, 14.01, 14.67, and 16.04 MeV, along with an assignment of $T = \frac{3}{2}$ for the 14.67 MeV state, though no low lying states could be resolved. [**Dietrich**]. The next work that found a potential $\frac{1}{2}+$ state was performed by K. Gul et al. Low lying states, such as the ground state, 2.345, 2.78, and 4.8 were identified, along with a peak between the ground state and the 2.33 state. However, this peak was attributed to a two-stage process of populating the 11.82 MeV state in ^9Be using $(^3He, p)$, followed by a neutron decay of ^9Be to ^8Be [**Gul**]. After this work, no other study would make an explicit attempt to measure the $\frac{1}{2}+$ state using the $(^3He, n)$ reaction. A measurement of the the $\frac{1}{2}+$ state in ^9B using the $(^3He, n)$ reaction with

modern detectors is sorely lacking.

The masses of nuclei within an isobaric multiplet are described by the Isobaric Multiplet Mass Equation (IMME):

$$M(T_z) = a(\alpha, T_z) + b(\alpha, T_z)T_z + c(\alpha, T_z)T_z^2$$

where α refers to any other relevant quantum numbers besides the isospin and the z-projection. This equation is derived by assuming the nucleons interact via the strong interaction, H_0, then assuming that the two-body Coulomb interactions between protons can be treated as a perturbation to H_0. This results in an equation that is quadratic in T_z, requiring at least $T = \frac{3}{2}$ isospin states to validate this equation. In order to deduce the coefficients to this equation, a fit to the mass excess for the nucleus in question is required, usually deduced from a measurement of Q-value [**PhysRevLett.25.530**]. A multiplet known to significantly deviate from this equation is the A = 9 $T = \frac{3}{2}$ quartet, involving ^9Li, ^9Be, ^9B, and ^9C.

The value of the energy of the first $\frac{1}{2}+$ state of ^9B is currently an unresolved issue within nuclear physics. After the first attempted measurement of this state in 1955 [**PhysRev.100.91**], there have been many attempts to measure this state with various reaction channels. Unlike its mirror nucleus ^9Be, the ground state of ^9B is unbound with a proton separation energy of -185.9 keV, meaning that all excited states exist as resonances. It lies in a region containing many broad resonances, making the task of measurement challenging.

CHAPTER 2

REACTION KINEMATICS

To populate states within nuclei, a commonly used method is to set up a reaction with known parameters that allows for the deduction of information from the populated nucleus. This involves a projectile nucleus, typically a light nucleus that has some energy with respect to the lab frame and a target nucleus that is at rest. The projectile interacts with the target, which results in a residual nucleus and an ejected nucleus, or ejectile, that travels outwards and is detected in spectroscopy. It is conventional to represent this type reaction as $A(a,b)B$, where a is the projectile, A is the target, B is the residual, and b is the ejectile. This reaction describes a transfer reaction, in which the interaction involves a transfer of nucleons between the projectile and target nucleus, and is the primary means used to populate states in ^9B for this project. In this case, the reaction is $^7Li(^3He,n)^9B$. In order to observe the residual nucleus created, a measurement of the resulting ejectile yields information based on kinematics. In our case, this ejectile is simply a neutron.

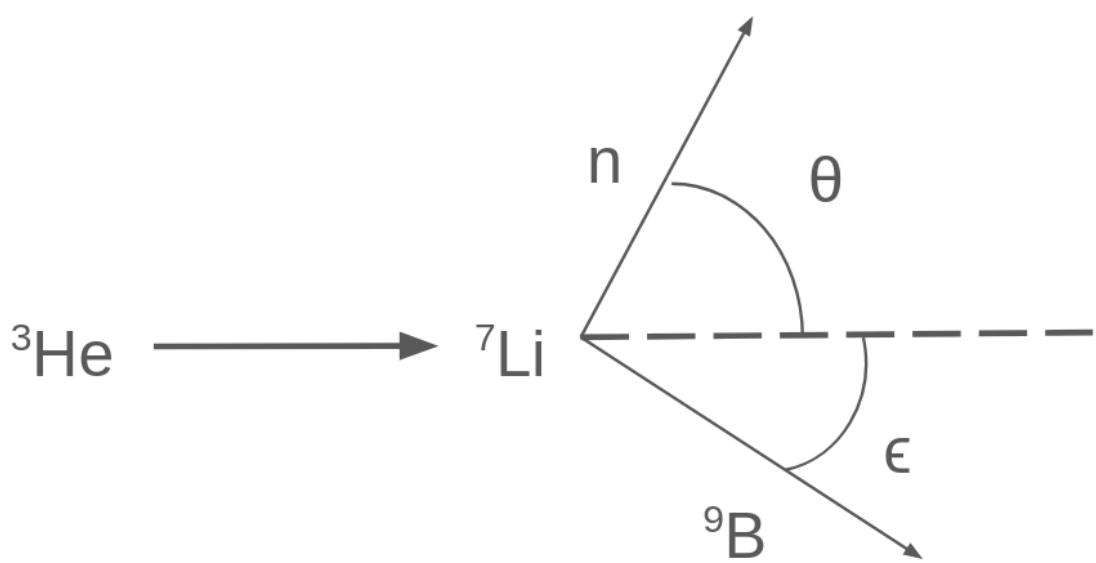

Figure 2.1: Kinematic representation of transfer reaction.

Neutron spectroscopy is the study of nuclear states from the measurement of neutron spectra. A technique for measuring this spectrum is time-of-flight (ToF) spectroscopy. ToF spectroscopy works by measuring the flight time of the neutron from the target to the detector. Since the neutron is a massive particle, its ToF is inversely proportional to its energy. In order to accurately determine the neutron flight time, the flight times of the prompt gammas emitted from excitations of the target from the beam are used. Since the gammas travel at the speed of light, a distinct peak can be identified within a ToF spectrum, allowing for the determination of neutron flight times. An advantage of this technique is the separation of neutrons from different excited states due to the dependence of ToF on neutron energy. This allows for the examination of all populated excitations of a given residual nucleus.

To derive the energies populated by the residual nucleus from the ejectile, one applies conservation of momentum p and energy E

$$p_a + p_A = p_b + p_B(1)E_a + E_A = E_b + E_B(2) \tag{2.1}$$

where $E = \frac{p^2}{2m} + mc^2$. The conservation of momentum can be written as

$$p_a = p_b cos\theta + p_B cos\epsilon(3)0 = p_b sin\theta + p_B sin\epsilon(4) \tag{2.2}$$

Using $p = \sqrt{2mT}$ with kinetic energy T and substituting (4) into (3), we have

$$T_B = \frac{1}{m_B}[m_a T_a + m_b T_b - 2\sqrt{m_a m_b T_a T_b} cos\theta$$

.

Going further, we can take the Q-value, or the difference in mass energy between the initial and

final nuclei, which is given as

$$Q = (m_a + m_A - m_b - m_B)c^2$$

Using (1) and rearranging with $E = T + \frac{p^2}{2m}$,

$$Q = (m_a - m_b - m_c)c^2 = T_b - T_a + T_b = T_b - T_a + \frac{1}{m_B}[m_a T_a + m_b T_b - 2\sqrt{m_a m_b T_a T_b} cos\theta$$

The excitation of the residual nucleus can then be expressed as

$$Q = Q_0 - E_{ex}(5)$$

,

where Q_0 is the Q-value of the ground state in ^9B and E_{ex} is the excitation energy of the nucleus. The excited states of ^9B is described by

$$E_{ex} = Q_0 - (m_a + m_A - m_b - m_B)c^2 = Q_0 - T_b - T_a + T_B = Q_0 - T_b - T_a + \frac{1}{m_B}[m_a T_a + m_b T_b - 2\sqrt{m_a m_b T_a T_b} cos\theta])$$

.

Regarding neutron interaction with scintillators, the primary means of energy transfer is via elastic scattering. The maximum amount of energy transferred can be parameterized as $E' = \frac{A-1}{A+1}E$. For deuterium, $A = 2$, so $E_{dep} = \frac{8}{9}E$. From this energy, the light-output can be parameterized as [**LAWRENCE201321**]

$L = aE_{dep} - b(1 - e^{-cE_{dep}})$

where a, b, and c are parameters that must be experimentally characterized. These coefficients have been determined via previous work. The table below is adapted from [**morelock2022characterization**].

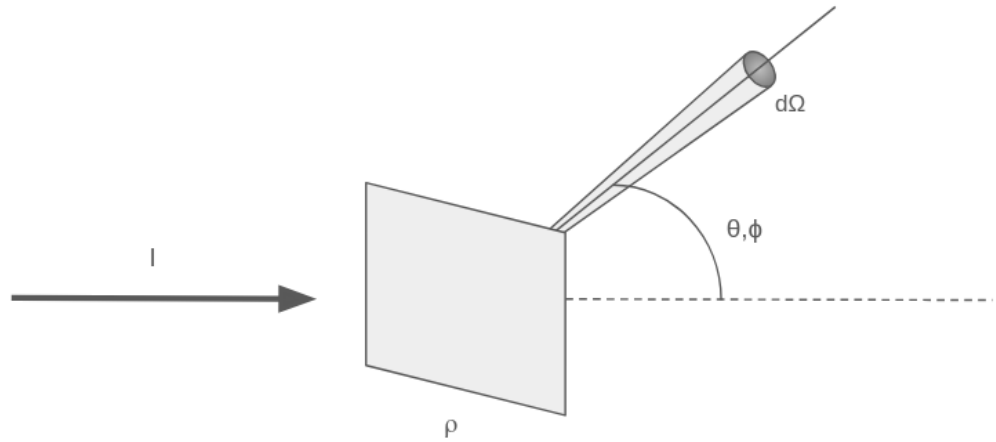

Figure 2.2: Visual example of kinematics in cross section calculation.

States in nuclei are further characterized by their total spin J and their parity Π. This J^{Π} value features within angular distribution plots that plot the differential cross section $\frac{d\sigma}{d\Omega}$ over the center of mass angle θ_{cm}. The differential cross section is a measure of the likelihood of a given reaction occurring as a function of the angular distribution of thereaction products. In order to compute this angular distribution, we use the equation

$\frac{d\sigma}{d\Omega} = \frac{N}{\epsilon \rho I d\Omega}$

where N is the number of ejectile nuclei detected, ϵ is the detector efficiency, ρ is the atomic density of the target, I is the number of projectile nuclei incident to the target, $d\Omega$ is the solid angle of the detector. These experimentally determined angular distributions can then be com-

pared to theoretical angular distributions using the Distorted Wave Born Approximation (DWBA) and optical model to find spectroscopic factors, denoted as C^2S. The experimental and theoretical angular distributions are related by

$$(\frac{d\sigma}{d\Omega})_{experiment} = C^2S(\frac{d\sigma}{d\Omega})_{theory}.$$

CHAPTER 3

CATRINA AND CHARON

3.0.1 CATRiNA

In order to detect particles such as a neutron, scintillators are used. Within the field of nuclear physics, it is customary to distinguish the type of neutron radiation based on the energy range of the observed neutrons. Fast neutrons are typically considered to be greater than 1 MeV in kinetic energy. Since the cross section for neutron interactions in most materials is strongly dependent on neutron energy, fast neutron detection relies on elastic scattering interactions. Scintillators function as a reliable way to determine neutron energy spectra. Scintillators refer to any material that emits light after interacting with radiation. This light is then converted into an electric signal via a Photomultiplier Tube (PMT).

The Compound Array for Transfer Reactions in Nuclear Astrophysics (CATRiNA) is an array of deuterated benzene scintillators. The scintillators are stored in a cylindrical container with two sizes: 4"x2" and 2"x2". The 4"x2" and 2"x2" scintillator is the EJ-315 deuterated liquid scintillator produced by Eljen Technology. The coupled Photomultiplier Tube (PMT) is the 9821B PMT developed by ET Enterprises for the 4"x2" scintillator and the Hamamatsu R7724 PMT for the 2"x2" scintillator. Both scintillators rely on the scintillation of deuterated benzene to accurately measure incoming neutron energies. The scintillating molecule, benzene, comprises a hexagon of carbon atoms with a deuterium atom bonded to each carbon atom. About two or three of the valence electrons of carbon occupy the σ orbital in between the carbon and deuterium atoms and are strongly localized. The remaining electrons occupy the π orbitals, which extend outside of the molecular plane and are not strongly localized [**BROOKS1979477**]. Transitions between the π orbital electrons are responsible for the luminescence seen within the deuterated benzene. A level diagram is provided in Figure. The electrons can be excited into either singlet or triplet states, denoted as S_i and T_i in Figure. Only transitions between levels emit gamma radiation. Transitions between vibrational states within the same level thermally de-excite. There are three main decays that can occur to excited π orbital electrons.

Prompt fluorescence refers to an electron, excited into a singlet state, decays into the ground state. This transition is short lived, with a half-life on the order of 1-80 ns. Phosphorescence refers to a process where a singlet electronic configuration is converted into a triplet configuration via inter-system crossing. Since the triplet has a much longer lifetime than the singlet with up to 10^{-3} s, this results in a delayed form of emission. Since the triplet state is at a lower energy relative to the singlet state, the emitted photon will have a longer wavelength than fluorescence photons. Delayed fluorescence refers to the process where a singlet state, converted into a triplet state, converts back into a singlet state before decaying via fluorescence [**knoll2010radiation**]. The intensity of the fluorescence decays can be represented by the $I = Ae^{-t/\tau_f} + Be^{-t/\tau_t}$, where τ_s and τ_f refers to the mean lifetimes of the slow and fast components of the emission spectrum (delayed fluorescence and fluorescence) and A and B corresponds to the relative magnitude of each component.

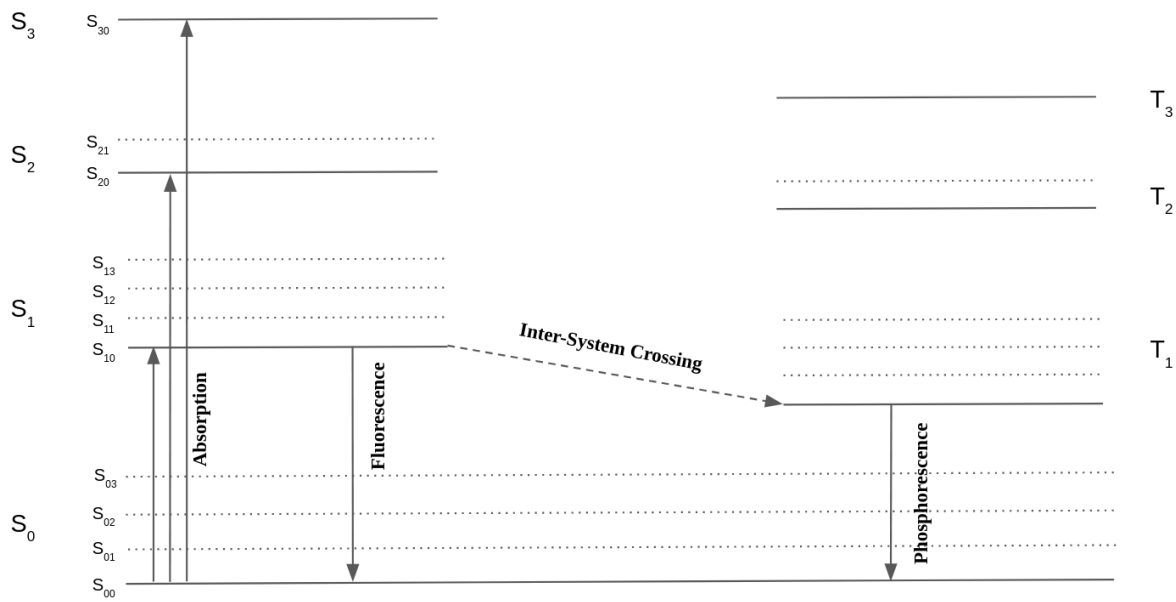

Figure 3.1: Organic Level Scheme

These two decays are basis of the Pulse-Shape Discrimination (PSD) technique for separating n-γ events. The relative magnitude of each component of the emission spectrum is dependent upon the type of incident radiation. γ radiation typically has a strong prompt fluorescence component but a small delayed fluorescence component. Neutron radiation has a smaller prompt fluorescence component but a larger delayed fluorescence component. This allows for the discrimination of these

events via long and short gates. As shown in Figure, the long gate is the total integral of the signal while the short gate is an integral over the rise time. The tail gate (long integral - short integral) can then be defined. This "tail integral" is consistently greater for neutron-induced events than γ-induced events, resulting in a PSD spectrum where each tail corresponds to a different radiation type.

Referring to the light-output incident radiation emits when interacting with scintillators, it is customary to define the unit electron-volt electron equivalent, or eVee. An eVee is defined as the light-output an electron with 1 ev of energy gives off. The PSD spectra obtained in this prospectus are plotted in terms of keVee and PSD ratio. The value of the long integral is defined in terms of arbitrary units, so a calibration is applied using ^{60}Co and ^{137}Cs. These two sources contain well-known Compton edges which allows for a linear calibration of the long integral into units of keVee.

3.0.2 CHARON

CHAPTER 4

CURRENT WORK

4.0.1 08/26/24

An experiment was run from 08/19/24 to 08/26/24 populating states in ^9B with the $^7Li(^3He,n)^9B$ reaction. This was achieved with the FN Tandem accelerator at the FSU John D. Fox lab. The ^3He beam has an energy of 7 MeV in the laboratory frame. This was the lowest possible beam energy the Tandem accelerator could coherently transmit the ^3He beam. This energy was minimized to reduce neutron energy so maximum separation could be achieved between different states. To record the data in a digitial format, CAEN digitizers were used to digitize signals. The logic setup is listed in the figure below.

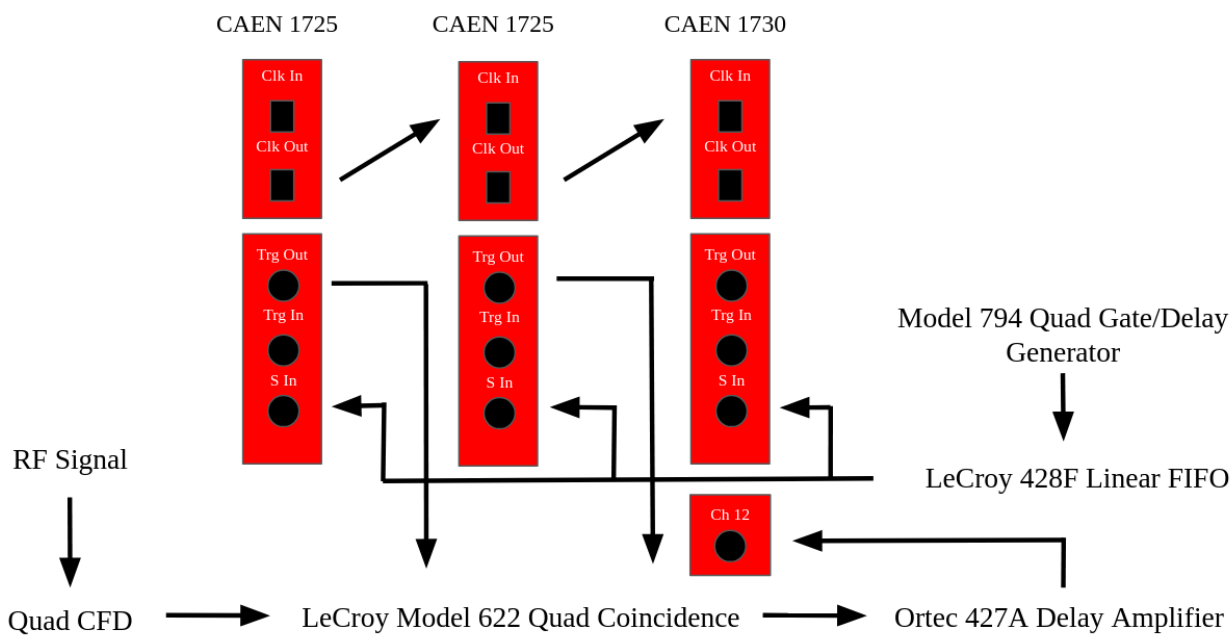

Figure 4.1: Cartoon showing logic setup used for the CAEN digitzers. The RF signal was recorded in the 1730 board.

The CATRiNA detectors were also placed at backwards angles to further reduce neutron energy. There were three neutron detectors placed at forward angles. All detectors except for the forward detectors were placed at 2 meters from the target ladder. This was done to ensure the anticipated